Chapter

Heat

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1. Heat capacity of a body is the:

(a) energy needed to melt a body without change in its temperature.

(b) energy needed to raise the temperature of a body by 1°C

(c) increase in volume of the body when its temperature increases by 1°C

(d) total amount of internal energy that is constant.

SQP 2025

2. The amount of heat energy required to melt a given mass of a substance at its melting point without any rise in its temperature is called as the:

(a) specific heat capacity

(b) specific latent heat of fusion

(c) latent heat of fusion

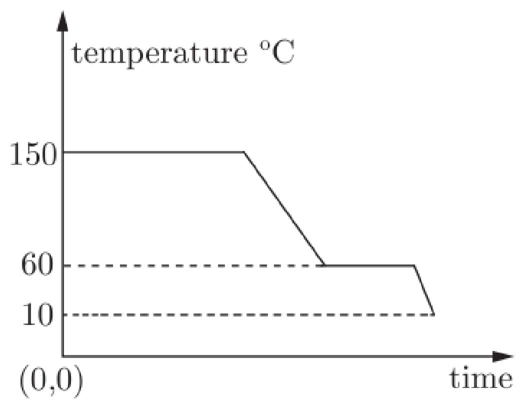
(d) specific latent heat of freezing

SQP 2025

3. The diagram below shows a cooling curve for a substance:

(a) State the temperatures at which the substance condenses.

(b) The temperature range in which the substance is in liquid state.



(c) Why do we prefer ice to ice-cold water for cooling a drink?

SQP 2025

4. A solid of mass 60 g at 100°C is placed in 150 g of water at 20°C. The final steady temperature is 25°C. Calculate the heat capacity of solid.

[sp. heat capacity of water = 4.2 J g^{-1} K^{-1}]

SQP 2025

5. During melting of ice at 0°C the

(a) energy is released and temperature remains constant.

(b) energy is absorbed and temperature remains constant.

(c) energy is released and temperature decreases.

(d) energy is absorbed and temperature increases.

MAIN 2024

6. Water freezes to form ice. What change would you expect in the average kinetic energy of the molecules?

MAIN 2024

7. Which will contain more heat energy 1 g of ice at 0°C or 1 g water at 0°C?

MAIN 2024

8. How much heat is required to convert 500 g of ice at 0°C to water at 0°C? The latent heat of fusion of ice is 330 Jg^{-1}.

MAIN 2024

9. 85 g of water at 30°C is cooled to 5°C by adding certain mass of ice. Find the mass of ice required. [Specific heat capacity of water = 4.2 J C^{-1}°C^{-1}, Specific latent heat of fusion = 336 J g^{-1}]

MAIN 2024

10. Explain why bottled soft drinks are more effectively cooled by cubes of ice than by ice water.

SQP 2023

11. The temperature recorded by a thermometer decreases when its bulb is covered with a piece of cloth soaked in spirit. Explain why.

SQP 2023

12. 50 g of a hot solid of specific heat capacity 0.25 J g^{-1} °C^{-1} and at 100°C is placed in 80 g of cold water, when the temperature of cold water rises by 3°C. Find the initial temperature of cold water.

(a) 6.36°C

(b) 26.36°C

(c) 16.36°C

(d) 36.36°C

MAIN 2023

13. Explain the following :

(i) Bottled drinks are cooled more effectively when surrounded by lumps of ice than iced water.

(ii) Why does atmospheric temperature fall after a hail storm?

(iii) Why does the weather become pleasant when it starts freezing in cold countries?

(iv) Why is water used as a coolant in motor car radiators?

SQP 2022

14. Explain the following :

(i) Why does the temperature in hot summer, falls sharply after a sharp shower?

(ii) Why do sandy soils, get heated up quickly as compared to wet soils?

(iii) Why is water considered as the best liquid for quenching thirst?

(iv) Why is it advisable to pour cold water over burns, caused on human body by hot solids?

SQP 2022

15. A hot solid of mass 60 g at 100°C is placed in 150 g of water at 20°C. The final steady temperature recorded is 25°C. Calculate the specific heat capacity of the solid. [Specific heat capacity of water 4200 J kg^{-1} °C^{-1}]

MAIN 2022

16. A vessel from which the air is rapidly being pumped out contains 100 g of water at 0°C. The intensive evaporation causes a gradual freezing of the water. What part of the original water can be converted into ice by this method ?

(a) 60%

(b) 78%

(c) 86.2%

(d) 56%

COMP 2022

17. Materials X, Y and Z are solids that are at their melting temperatures. Material X requires 200 J to melt 4 kg, Y requires 300 J to melt 5 kg and Z requires 300 J to melt 6 kg. Rank the materials according to their heats of fusion in descending order.

COMP 2022

18. Calculate the amount of ice which is required to cool 150 g of water contained in a vessel of mass 100 g at 30°C, such that the final temperature of the mixture is 5°C. (Take specific heat capacity of material of vessel as 0.4 J g^{-1} $°C^{-1}$, specific latent heat of fusion of ice = 336 J g^{-1}, specific heat capacity of water 4.2. J g^{-1} $°C^{-1}$).

COMP 2022

19. A liquid of mass 100 g loses heat at a rate of 200 Js^{-1} for 1 minute. If the temperature of liquid drops by 100°C, Calculate the specific heat capacity of the liquid.

(a) 0.6 Jg^{-1} $°C^{-1}$

(b) 1.2 Jg^{-1} $°C^{-1}$

(c) 0.8 Jg^{-1} $°C^{-1}$

(d) 1.6 Jg^{-1} $°C^{-1}$

SQP 2021

20. A heater, rated 1000 W is used to heat 1.5 kg of water at 40°C to its boiling point. Calculate the time in which the water starts to boil. Specific heat capacity of water is 4200 J kg^{-1} $°C^{-1}$.

(a) 378 seconds

(b) 225 seconds

(c) 480 seconds

(d) 680 seconds

MAIN 2021

21. A strong sensation of cooling is produced when a little methylated spirit is applied on the hand. Explain why. Suggest one way by which the cooling sensation may be increased.

COMP 2021

Solution

1.

Ans: (b) energy needed to raise the temperature of a body by 1°C

Explanation:

Heat capacity of a body is defined as the **amount of heat energy required to raise the temperature of the whole body by 1°C** (or 1 K).

So, option **(b)** is correct.

2.

Ans: (c) latent heat of fusion

Explanation:

The heat energy required to **convert a solid into a liquid at its melting point without any change in temperature** is called the **latent heat of fusion**.

Therefore, the correct option is **(c)**.

3.

Ans:

(a) The substance condenses at **150°C**, where the temperature remains constant during cooling.

(b) The substance is in the liquid state in the temperature range **150°C to 60°C**.

(c) Ice is preferred over ice-cold water because **ice absorbs much more heat (336 J per gram) while melting**, due to its high latent heat of fusion. Therefore, ice cools a drink more effectively than ice-cold water.

4.

Ans: By the **principle of mixtures**:

Heat lost by solid = Heat gained by water

For the solid:

- Mass m = 60 g
- Initial temperature =100° C

- Final temperature = 25° C
- Fall in temperature ΔT = 100 − 25 = 75° C

For water:

- Mass = 150 g
- Rise in temperature ΔT = 25 – 20 = 5° C
- Specific heat capacity = 4.2 J g^{-1} K^{-1}

Using the formula:

$$(m\,c\,\Delta T)_{\text{solid}} = (m\,c\,\Delta T)_{\text{water}}$$

$$60\,c' \times 75 = 150 \times 4.2 \times 5$$

$$c' \times 75 = 150 \times 21$$

$$c' \times 75 = 3150$$

$$c' = 42\,\mathrm{J\,K^{-1}}$$

Heat capacity of the solid = 42 J K^{-1}

5.

Ans: (b) energy is absorbed and temperature remains constant

Explanation:

When ice melts at 0°C, it **takes in heat energy** from the surroundings. This heat is used to **change the state** of ice into water (latent heat of fusion), not to raise its temperature. So, the temperature stays at **0°C** until all the ice has melted, even though energy is being absorbed.

6.

Ans: Since there is no change in temperature during the freezing process, the average kinetic energy of the molecules remains unchanged.

7.

Ans: 1 g of ice (solid) at 0°C contains extra heat in the form of its latent heat of fusion, whereas 1 g of water (liquid) at 0°C has no latent heat.

Hence, **1 g of ice at 0°C contains more heat energy than 1 g of water at 0°C.**

8.

Ans:

Required heat,

$$Q = mL$$

$$Q = 500 \times 330 = 165000 \text{ J}$$

9.

Ans:

Let the mass of ice added be m g.

Heat lost by water:

$$= 85 \times 4.2 \times (30 - 5)$$

Heat gained by ice:

$$= m \times 336 + m \times 4.2 \times (5 - 0)$$

By the principle of calorimetry:

$$85 \times 4.2 \times (30 - 5) = m \times 336 + m \times 4.2 \times 5$$

$$8925 = 357\, m$$

$$m = 25 \text{ g}$$

10.

Ans: Both ice cubes and ice-cold water are at 0∘C0∘C, but ice cubes cool better because each gram of ice needs **extra heat (80 calories)** to melt and become water at the same temperature. This extra heat required for melting increases the cooling effect.

Hence, the **cooling capacity of ice cubes is greater** than that of ice water.

11.

Ans: When spirit evaporates, it requires latent heat of vaporisation to change into vapour. This heat is taken from the surroundings, including the mercury in the bulb of the thermometer. As heat is removed from the thermometer bulb, its temperature falls.

Therefore, the temperature recorded by the thermometer decreases.

12.

Ans:

Given:

Mass of hot solid,

$$m_1 = 50\text{ g},\quad c_1 = 0.25\text{ J g}^{-1}{}^\circ\text{C}^{-1},\quad T_1 = 100^\circ\text{C}$$

Mass of cold water,

$$m_2 = 80\text{ g},\quad c_2 = 4.2\text{ J g}^{-1}{}^\circ\text{C}^{-1}$$

Rise in temperature of water:

$$\Delta T = 3^\circ\text{C}$$

Let the initial temperature of cold water be θ °C.

Final temperature of mixture = $\theta + 3^\circ$C

By the **principle of mixtures**:

$$\text{Heat lost by hot solid} = \text{Heat gained by water}$$

$$m_1c_1(100 - (\theta + 3)) = m_2c_2(3)$$

$$50 \times 0.25 \times (97 - \theta) = 80 \times 4.2 \times 3$$

$$12.5\,(97 - \theta) = 1008$$

$$1212.5 - 12.5\,\theta = 1008$$

$$12.5\,\theta = 204.5$$

$$\theta = 16.36^\circ\text{C}$$

Thus, the **initial temperature of cold water is**:

$$\boxed{16.36^\circ\text{C}}$$

Correct option: **(c) 16.36°C**

13.

Ans:

(i) Lumps of ice cool bottled drinks more effectively than iced water because **each gram of ice at 0° C** absorbs an additional **336 J** of heat (latent heat of fusion) to melt into water at the same temperature. This extra heat absorption increases the cooling effect, making ice more effective than ice-cold water.

(ii) During a hailstorm, small pieces of ice fall and begin to melt. As they melt, they absorb a **large amount of heat from the surrounding air**, because ice has a very high latent heat of fusion. This removal of heat from the atmosphere lowers the atmospheric temperature, making it fall rapidly.

(iii) When water starts freezing in cold countries, each kilogram of water at **0° C releases 336,000 J** of heat energy (latent heat of fusion). This heat is given out into the surroundings. The release of such a large amount of heat makes the weather feel **more pleasant** despite the low temperature.

(iv) Water is used as a coolant because it has a **very high specific heat capacity ($4.2\ J\ g^{-1}°C^{-1}$)**. This means it can absorb a large amount of heat without a significant rise in temperature. As water absorbs heat from the engine, it prevents overheating, and the absorbed heat is later released through the radiator—thus controlling the engine temperature efficiently.

14.

Ans:

(i) Water has a very high specific heat capacity of **$4.2\ J\ g^{-1}\ °C^{-1}$**. During a sharp shower, raindrops evaporate and absorb a large amount of heat from the surroundings. As this heat is taken from the air, the atmospheric temperature **falls sharply**.

(ii) The specific heat capacity of sand is about **five times lower** than that of water. Therefore, sand requires much less heat to raise its temperature, while wet soil (containing water) needs more heat to warm up. Thus, **sandy soil heats up faster** than wet soil.

(iii) Water has a very high specific heat capacity, so it can absorb a large amount of heat without a significant rise in temperature. When we feel thirsty, it is a signal that the body has produced more heat than usual. Water absorbs this excess heat effectively, making it the **best liquid for quenching thirst**.

(iv) Because water has a high specific heat capacity, it can absorb a **large amount of heat rapidly**. When cold water is poured over a burn, it quickly removes heat from the affected area, thereby **providing fast relief** and preventing further skin damage.

15.

Ans:

Let the specific heat capacity of the solid be c.

According to the **principle of calorimetry**:

Heat lost by the hot solid = Heat gained by water

For the solid:

m = 60 g, ΔT = (100 − 25) **°C** =75**°C**

For the water:

$$m = 150\,\text{g}, \quad c = 4.2\,\text{J g}^{-1\circ}\text{C}^{-1}, \quad \Delta T = (25 - 20)^\circ\text{C} = 5^\circ\text{C}$$

Applying the formula:

$$60 \times c \times 75 = 150 \times 4.2 \times 5$$

$$c = \frac{150 \times 4.2 \times 5}{60 \times 75}$$

$$c = 0.7\ \text{J g}^{-1\circ}\text{C}^{-1}$$

Thus, the **specific heat capacity of the solid is**:

$$\boxed{0.7\ \text{J g}^{-1\circ}\text{C}^{-1}}$$

16.

Ans: (c) 86.2%

Explanation:

When air is pumped out rapidly, water at 0°C evaporates. The heat required for this evaporation is taken from the remaining water, causing it to freeze.

- **Latent heat of vaporisation at 0°C:**

  $$L_v = 2.10 \times 10^6\ \text{J/kg} = 2.10 \times 10^3\ \text{J/g}$$

- **Latent heat of fusion of ice:**

  $$L_f = 3.36 \times 10^5\ \text{J/kg} = 3.36 \times 10^2\ \text{J/g}$$

Let the mass of ice formed = m g.

Then mass of water evaporated = (100 − m) g.

Since **heat lost in evaporation = heat gained to freeze water**:

$$(100 - m)L_v = mL_f$$

$$(100 - m)(2.10 \times 10^3) = m(3.36 \times 10^2)$$

Solving:

$$210000 - 2100m = 336m$$

$$210000 = 2436m$$

$$m \approx 86.2 \text{ g}$$

Thus, **86.2% of the water turns into ice**, which matches **option (c)**.

17.

Ans:

The heat of fusion is given by:

$$L = \frac{Q}{m}$$

For material X:

$$L_X = \frac{200 \text{ J}}{4 \text{ kg}} = 50 \text{ J/kg}$$

For material Y:

$$L_Y = \frac{300 \text{ J}}{5 \text{ kg}} = 60 \text{ J/kg}$$

For material Z:

$$L_Z = \frac{300 \text{ J}}{6 \text{ kg}} = 50 \text{ J/kg}$$

Thus, the order of heat of fusion in **descending order** is:

$$\boxed{L_Y > L_X = L_Z}$$

18.

Ans: According to the principle of mixtures,

Heat lost by water and vessel = Heat gained by ice to melt and warm to 5°C

Heat lost by water:

$$150 \times 4.2 \times (30 - 5)$$

Heat lost by vessel:

$$100 \times 0.4 \times (30 - 5)$$

Total heat lost:

$$150 \times 4.2 \times 25 + 100 \times 0.4 \times 25$$

Heat gained by ice:

$$m \times 336 + m \times 4.2 \times (5 - 0)$$

Now equating:

$$150 \times 4.2 \times 25 + 100 \times 0.4 \times 25 = m(336 + 4.2 \times 5)$$

$$m = 46.9 \text{ g} \approx 47 \text{ g}$$

19.

Ans: (b) 1.2 J g^{-1} $°C^{-1}$

Explanation:

- Mass of liquid, $m = 100 \text{ g}$
- Heat lost in 1 minute $= 200 \text{ J s}^{-1} \times 60 \text{ s} = 12000 \text{ J}$
- Fall in temperature, $\Delta T = 100^{\circ}\text{C}$

Using $H = mc\Delta T$:

$$c = \frac{H}{m\Delta T} = \frac{12000}{100 \times 100} = 1.2 \text{ J g}^{-1\circ}\text{C}^{-1}$$

So, the specific heat capacity of the liquid is **1.2 J g^{-1} $°C^{-1}$**.

20.

Ans: (a) 378 seconds

Explanation:

To raise the temperature of 1.5 kg of water from 40°C to 100°C, heat required is:

$$H = mc\Delta T$$

$$H = 1.5 \times 4200 \times 60 = 378000 \text{ J}$$

The heater's power is 1000 W, so:

$$t = \frac{H}{P} = \frac{378000}{1000} = 378 \text{ seconds}$$

Hence, the water starts boiling after **378 seconds**.

21.

Ans: Methylated spirit is very volatile, so it evaporates quickly. During evaporation, it absorbs latent heat from the hand. This loss of heat from the hand produces a strong cooling sensation.

The cooling effect can be increased by **blowing air** over the hand, which speeds up evaporation.